Total No. of Printed Pages:03

SUBJECT CODE NO: - RNP-8044
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - II
Examination May/June-2026
Applied Mathematics-II

[Time: 2:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION – A

Q.1 Solve any 10 of the following questions: **20**

1) Skewness of a normal distribution =
   a) 0 b) +1 c) -1 d) 3.

2) Mean deviation is the mean of:
   a) Deviations b) Absolute deviations
   c) Squared deviations d) None.

3) The square of SD gives:
   a) Mean b) Variance
   c) Median d) Mode.

4) Bowley's skewness uses:
   a) Mean b) Median
   c) Quartiles d) Mode.

5) The value $\int_0^{\pi/2} sin^4\theta cos^3\theta . d\theta$ is
   a) 0 b) 1 c) $\frac{2}{35}$ d) $\frac{35}{2}$

6) The value of $\int_0^{\pi} e^{-x} x^{-1/2} dx$ is
   a) $\sqrt{\pi}$ b) π c) 0 d) 1

7) The expression $\sqrt{1+\left(\frac{dy}{dx}\right)^2}$ is used in rectification of curves in:
   a) Polar form b) Cartesian form
   c) Parametric form d) None of the above.

8) The polar curve $r^2 = a^2 \cos 2\theta$ represents
   a) Circle b) Cardioid c) Lemniscate d) Spiral.

9) Volume under surface $z = f(x, y)$:
   a) $\iiint 1\, dV$ b) $\iint f(x, y)\, dA$
   c) $\int f(z)\, dz$ d) None.

10) Find the value of $\iint xydxdy$ over the area bounded by parabola $y = x^2$ and $x = -y^2$ is?
   a) 1/67   b) 1/24   c) -1/6   d) -1/12.

11) Surface generated by revolution:
   a) $2\pi\int yds$ b) $2\pi\int xds$
   c) $2\pi\int rds$ d) All.

12) The value of $\int_0^\infty\int_0^\infty \frac{dxdy}{(1+x^2)(1+y^2)}$ is
   a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{\pi^2}{4}$ d) $\frac{\pi^2}{8}$

13) For the data 2, 4, 6, 8, 10, the mean is:
   a) 4   b) 5   c) 6   d) 8.

SECTION – B

Q.2 a) Find Karl-Pearson's coefficient of skewness for the given distribution: **05**

X:	0–5	5–10	10–15	15–20	20–25	25–30	30–35	35–40
F:	2	5	7	13	21	16	8	3

b) Calculate mean deviation about mean: **05**

C.I	0-10	10-20	20-30	30-40	40-50	50-60	60-70
f	6	5	8	15	7	6	3

Q.3 a) Prove that $\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx = B(m,n)$ **05**

b) Evaluate $\int_0^{2a} x\sqrt{2a-x}dx$ **05**

Q.4 a) Show that $\int_0^1\int_0^{\sqrt{1+x^2}} \frac{dxdy}{1+x^2+y^2} = \frac{\pi}{4}[log(1+\sqrt{2})]$. **05**

b) Evaluate $\iint ydxdy$ over the area $y = x^2$ and $x + y = 2$ **05**

Q.5 a) Trace the curve $a^2y^2 = x^2(a^2 - x^2)$ with justification **05**

b) Find the length of the cardioid $r = a(1 + cos\,\theta)$ which lies outside the circle $r = acos\theta$. **05**

Q.6 a) Find mean mode & median for the following data: **05**

Age (in years)	10–19	20–29	30–39	40–49	50–59	60–69	70–79
Frequency	03	61	223	137	53	19	04

b) Find variance and coefficient of Variance: **05**

Marks	0–10	10–20	20–30	30–40	40–50	50–60
No. of Students	10	20	15	25	30	25

Q.7 a) Evaluate $\iiint \frac{dxdydz}{(1+x+y+z)^3}$ over the volume of tetrahedron bounded by $x = 0$, $y = 0, z = 0$ and x + y + z = 1 **05**

b) Change to polar co-ordinate & evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{xdxdy}{\sqrt{x^2+y^2}}$ **05**